

3 Fig. 3.1 shows the nematode worm, *Caenorhabditis elegans*.



Fig. 3.1

Stem cells of *C. elegans* have been studied.

Fig. 3.2 shows the change in mass of DNA per nucleus in a stem cell during one cell cycle.

$$1 \text{ pg} = 1 \times 10^{-12} \text{ g}$$

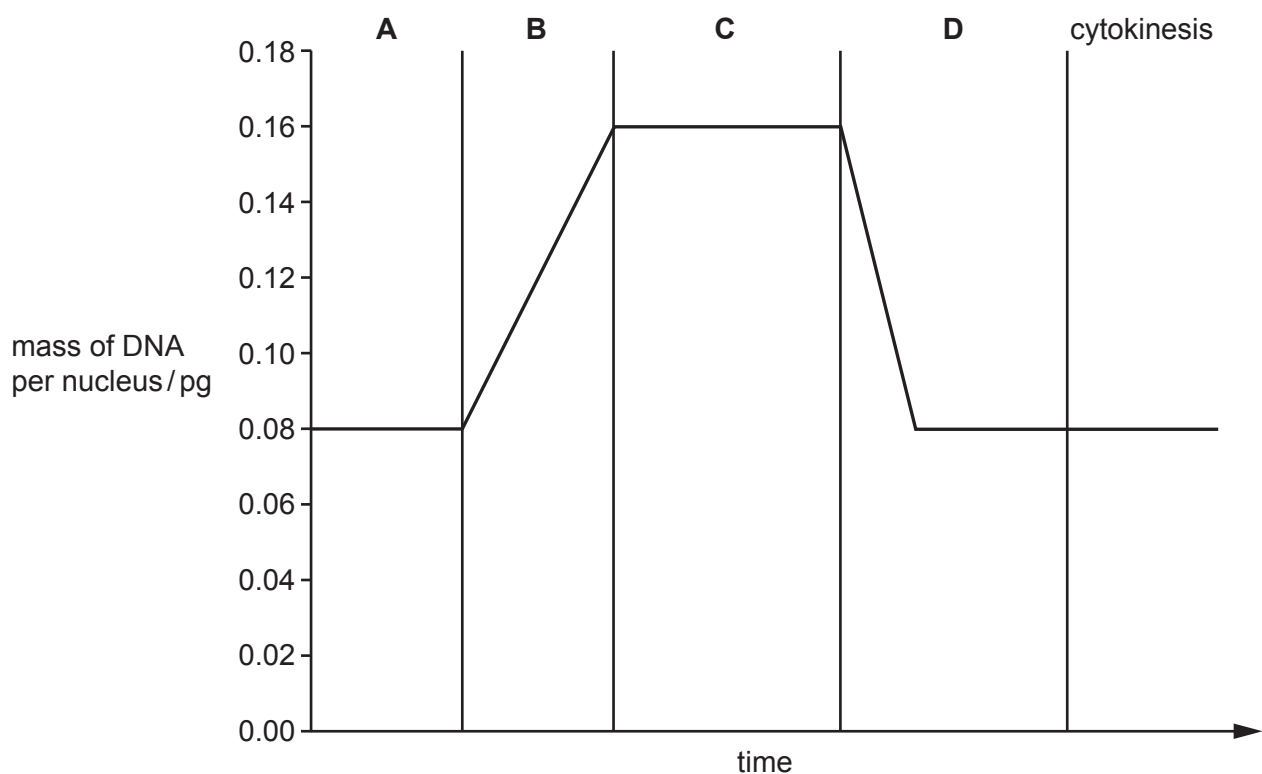


Fig. 3.2

(a) (i) State the phases of interphase shown by **A** and **B** in Fig. 3.2.

A

B

(ii) State the stage of mitosis shown by **D** in Fig. 3.2.
 [1]

(iii) Outline what happens in a cell in preparation for cytokinesis.

 [2]

(iv) A stem cell of a female *C. elegans* has 12 chromosomes.
 Complete Table 3.1 to show the number of nuclei within the stem cell in stages **A**, **B** and **D** of the cell cycle shown in Fig. 3.2 **and** the number of chromosomes in each nucleus during these stages.

Table 3.1

	stage of the cell cycle		
	A	B	D
number of nuclei within the stem cell			
number of chromosomes in each nucleus			

[2]

(b) Young nematodes have stem cells throughout the body.

Fig. 3.3 summarises three cell cycles of one of these stem cells.

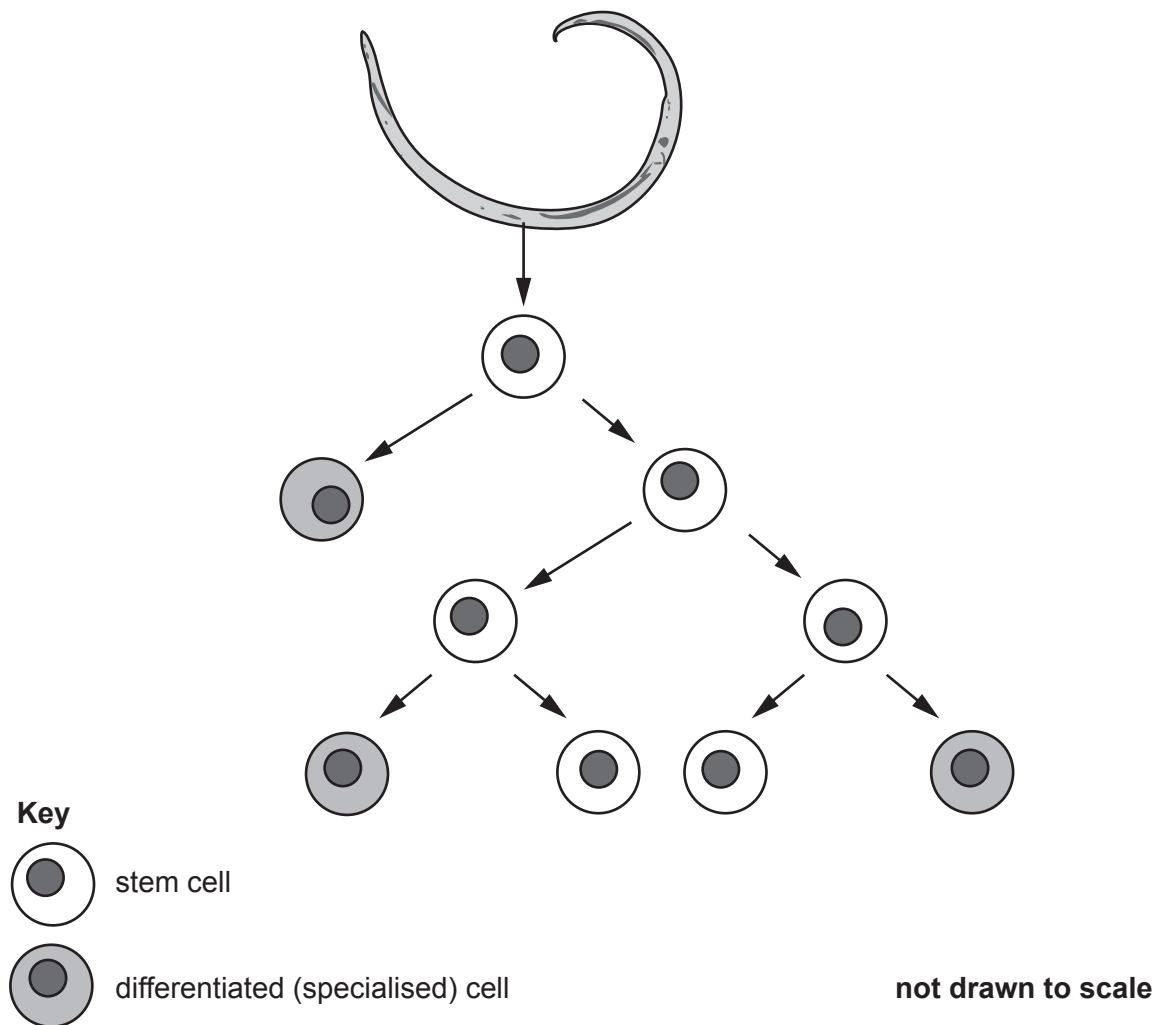


Fig. 3.3

With reference to Fig. 3.3, outline the role of stem cells in animals.

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